

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Covalent Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

A–E

1. A shared pair of electrons attracted to both nuclei.
2. An outer-shell pair not used in bonding.
3. e.g. H_2O or CO_2 ; diamond, graphite or SiO_2 .
4. Each Cl shares one electron; the shared pair is attracted to both nuclei.
5. Each C bonded to four others in a tetrahedral giant covalent network.
6. Graphite has delocalised electrons between layers; diamond's electrons are all in C–C bonds.
7. 1, 2, 3 and 4 shared pairs.
8. Water is simple molecular (weak intermolecular forces). SiO_2 is giant covalent.
9. Boiling overcomes intermolecular forces; covalent bonds inside molecules remain.
10. Each H has 1 electron. They share a pair. Both then have a full first shell. [3]
11. Iodine is simple molecular — weak forces between I_2 . Diamond is giant covalent. [4]
12. No mobile ions or delocalised electrons. [2]
13. Nitrogen donates its lone pair to H^+ (dative bond). All four N–H bonds in NH_4^+ are then equivalent.